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Lab task # 4

Object Oriented Programming

Question: 1

◆ Write a Java Program to Create a Calculator.

✧ **Input :**

```
package com.mycompany.exe;
import java.util.Scanner;

public class Exe {
    public static void main(String[] args) {
        int a, b, res;

        Scanner s = new Scanner(System.in);

        System.out.println("Enter values:");

        a = s.nextInt();
        b = s.nextInt();

        Cal1 c1 = new Cal1();
        Cal2 c2 = new Cal2();

        res = c1.sum(a, b);

        System.out.println("The sum of " + a + " and " + b + " is: " + res);

        c1.mul(a, b);

        System.out.println("Division is: " + c2.div(a, b));

        int y = c2.mod(a, b);

        System.out.println("Remainder is " + y);

    }
}
```

Main Class

Class 1

```
package com.mycompany.exe;
import java.util.Scanner;

public class Cal1 {

    public int sum(int a, int b) {

        int res = a + b;

        return res;

    }

    public void mul(int a, int b) {

        int res = a * b;

        System.out.println(res);

    }

}
```

Class 2

```
package com.mycompany.exe;
import java.util.Scanner;

public class Cal2 {

    public int div(int a, int b) {

        int res = a / b;

        return res;

    }

    public int mod(int a, int b) {

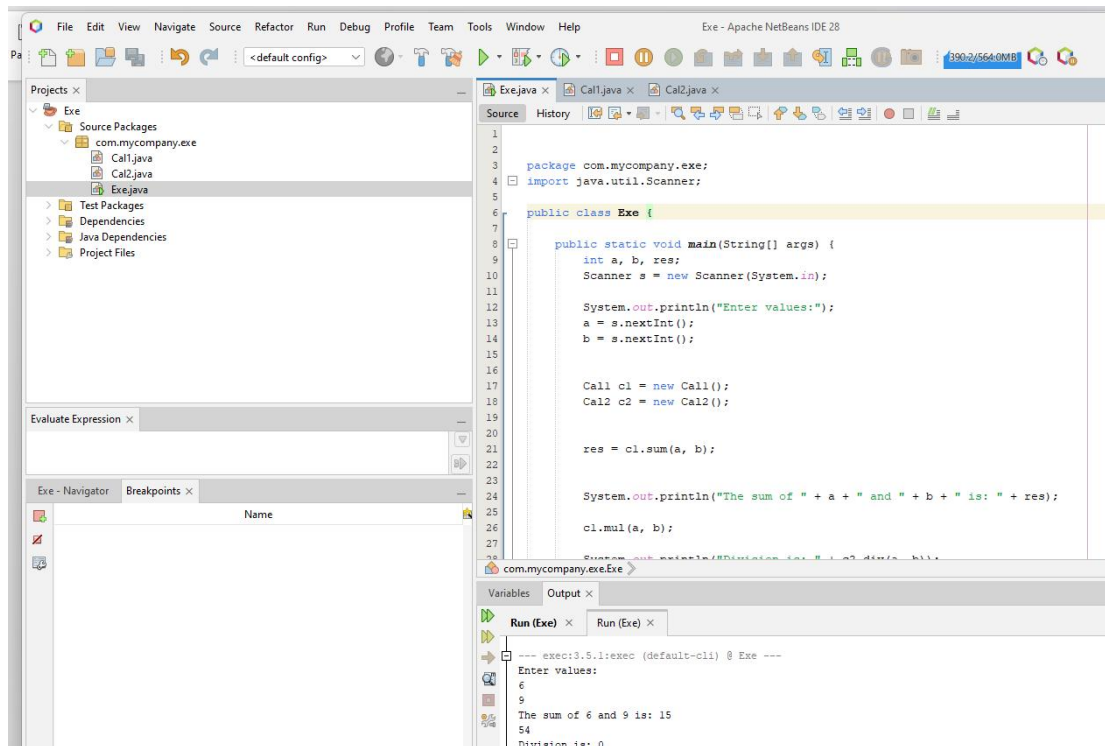
        int result = a % b;

        return result;

    }

}
```

✧ Output:



◆ **Question 2:** Make animal class and 2 other classes have name cats and dogs. Call cats and dogs feature in animal class.

✧ Input:

```
package com.mycompany.animal;
```

```
public class Animal {
```

```
    public void sound() {
```

```
        System.out.println("Animal makes a sound"); }
```

```
    public static void main(String[] args) {
```

```
        Animal myDog = new Dog();
```

```
        Animal myCat = new Cat();
```

```
        System.out.println("Testing Dog");
```

```
        myDog.sound();
```

```
        System.out.println("\n--- Testing Cat ---");
```

```
        myCat.sound(); }
```

```
}
```

Animal Class

Cat Class

```
package com.mycompany.animal;  
  
public class Cat extends Animal {  
  
    public void sound() {  
        System.out.println("Cat meows: Meow Meow!");  
    }  
}
```

DOG Class

```
package com.mycompany.animal;  
  
public class Dog extends Animal {  
  
    public void sound() {  
        System.out.println("Dog barks: Woof Woof!");  
    }  
}
```

Output:

Testing Dog

Dog barks: Woof Woof!

--- Testing Cat ---

Cat meows: Meow Meow!